SON 2355: Intelligent National Characterization of Harmful Algal Blooms Using Integrated Observations

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Problem:

USACE lacks consistent, long-term capabilities to characterize Harmful Algal Bloom (HAB) occurrence in reservoir systems. Existing knowledge is fragmented across site-specific studies, and satellite-derived HAB observations have not been connected to the process-based water quality models that Districts use for reservoir operations and planning. USACE cannot systematically compare bloom behavior across projects, identify long-term trends, or prioritize monitoring resources at the portfolio level.

Envisioned Product:

An AI-enabled geospatial analysis framework applied to 50 priority USACE reservoirs, producing a quality-assured database of historical HAB occurrence, frequency, intensity, and spatial extent. Deliverables will include validated ML classification workflows, CE-QUAL-W2 integration tools for using satellite-derived observations as model inputs and calibration targets, an ERDC technical report, and open-source software. The framework will be designed to scale to the full Civil Works portfolio.

Value Added:

This project will produce a consistent, multi-decadal baseline of historical HAB occurrence for USACE reservoirs by fusing time series of satellite imagery with hydrologic and meteorological observations through machine learning. By coupling the satellite-derived characterization with the CE-QUAL-W2 water quality model, the project connects data-driven detection with the process-based modeling tools Districts already use, enabling improved bloom simulation and scenario evaluation for reservoir operations and planning.

Regional/National Significance:

Framework and database will be applicable across all USACE Districts and Divisions. The 50-reservoir subset will span diverse hydrologic regimes, geographic regions, and reservoir types to ensure national representativeness. The workflows will utilize Google Earth Engine and its datasets, combined with other environmental geospatial and time series data. The products will link with with the interagency Cyanobacteria Assessment Network (CyAN) data infrastructure.